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Application for the Ph.D. position in the HypoWaves project — Bio 01/2026

Dear Dr Henkel, dear Dr Ahmerkamp,

Advective porewater flow turns permeable sand into a biocatalytic filter, and the paper Fabian Lange and Dr Ahmerkamp published this year (Lange et al. 2026) gives the organic matter in that filter a dual role: the glycans drive the oxygen consumption and, by stabilising the bedforms, sustain the advective supply of oxygen that feeds it. The study hypothesises bedform stability as a key control on remineralisation and fixed-nitrogen loss, then measures the carbon and the oxygen; the nitrogen rates are still unmeasured. That is the measurement I would like to make.

The pelagic half is open in your own words: the conditions under which nitrification or denitrification becomes the most important N_2O pathway in coastal areas *“have yet to be identified”* (Voss et al. 2025). Wan and colleagues (2023) showed that even in oxygenated coastal water the reductive pathway supplies most of the N_2O and collapses once the particles are filtered out — but the particle enters their analysis only as bulk quantity: no organic composition, no lability. So the question I would test is whether what a Baltic aggregate releases as N_2O is set by how fast oxygen falls and returns rather than by how low it goes, and whether the aggregate’s own organic matter decides which. Beside the ^{15}N rates I would put a compound-specific carbon measurement on the particle’s lipids. It fails cleanly: if the partitioning tracks the oxygen minimum alone, the idea is wrong and should be reported so.

Dr Ahmerkamp, in the Janssand work with Hannah Marchant (Marchant et al. 2018) you found no strong transcriptional response across the tidal cycle and concluded that community composition rather than regulation drives the imbalance in the denitrification intermediates; I have wondered since whether rapid, repeated oxygen swings would be the forcing under which selection and regulation part company. Dr Henkel, your *Sulfurimonas* work (Henkel et al. 2022) taught me the opposite discipline: isolate the organism, measure the rate per cell, and only then scale it. That is the argument I would want to be able to make for N_2O.

Transport is the line through my own work. My master’s thesis ran across MARUM, the Max Planck Institute for Marine Microbiology and the University of Bremen within TRR 420 — a geoscientist’s question answered in a glycobiology laboratory: I followed the degradation of model sinking particles by marine *Vibrio* strains image by image, and carried the result through a settling model to the exponent of the Martin curve. On Helgoland Mud Area sediments I designed ^{13}C-acetate incubations, followed the label into single fatty acids by GC-C-IRMS, and read porewater fluxes with Fick’s law. Comparing that core with itself weeks later gave me an unintended record of what happens when oxygen reaches reduced sediment — sulfide oxidised away, iron and manganese re-precipitating, phosphate re-adsorbed. HypoWaves asks that question on purpose.

From the GLOMAR stable-isotope course I know the ^{15}N isotope-pairing framework — the $^{29}N_2$:$^{30}N_2$ distribution, the exetainer time series, the pool-dilution and bottle-effect corrections, membrane-inlet mass spectrometry. What I have not done is run one on a real sample, and I would rather say so than dress lipid chemistry up as molecular ecology.

My first experiment is concrete: Baltic aggregates in suspension, the oxygen driven between saturation and hypoxia by mass-flow-controlled gas mixing — falling fast, falling slowly, held steady as controls — ^{15}N-nitrate added, N_2 by isotope pairing on the MIMS, N_2O by gas chromatography, the carbon arm beside it. IOW is where I want to do it: the flume, the landers, the stable-isotope and nanoSIMS facilities, the physics department, and a vessel with the draught for a genuinely shallow coast, all in one house. Having co-organised a two-week campaign for more than twenty students in the Argentinian Puna, helping to run the field campaigns is work I would take on gladly.

I defended my M.Sc. thesis on 23 March 2026 and am free to begin from 1 November 2026. Reading the two of you side by side is what decided me: the transport physics and the redox microbiology have not yet been joined on a particle, and this is the project where that happens.

Yours sincerely,

Osman Can Kandemiroglu